AI-Assisted EV Range Prediction and Smart Charging Optimization System

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ABSTRACT—Electric vehicle trip planning faces challenges to uncertainty in energy use, battery levels, road conditions and charging needs. This paper introduces an AI-driven system for predicting EV range and optimizing charging. The system combines route planning, machine learning for energy prediction, battery state tracking and charging station selection using mathematical optimization. It works within the Chennai District using browser-based tools for location input, like geolocation address search and map selection. A road network built from OpenStreetMap data enables route calculations. Dijkstra's algorithm finds the path with alternate routes available for users to choose. An XGBoost regression model predicts energy use based on vehicle details, battery condition, road type, speed, weather and driving behavior. The model's output is used with the starting battery state of charge to calculate energy, driving range and arrival SOC ensuring a minimum 25% battery reserve. If the trip exceeds the range the system identifies nearby charging stations that are reachable. It then uses a PuLP-based mixed-integer linear programming model to decide when and where to charge. The system is built with React, MapLibre, FastAPI, MongoDB, Python, OSMnx, NetworkX, XGBoost and PuLP. Testing with 20,000 real EV trip records showed performance: XGBoost achieved an R^2 of 0.9978, a MAE of 0.1481 and RMSE of 0.3613

INDEX TERMS—Electric Vehicles, Energy Consumption Prediction, XGBoost State of Charge Driving Range, Dijkstra Algorithm, Charging Station Selection, Smart Charging, Mixed-Integer Linear Programming, PuLP, Route Planning, Geospatial Computing.

I. INTRODUCTION

A. BACKGROUND

Trip planning for an electric vehicle is influenced by the fluctuations in energy consumption, state of charge of the battery, driving environment, road features and charging infrastructure. An optimal route will depend not only on the distance but also on the amount of estimated energy consumption and the battery charge left. A lot of work has been done in the past on predicting the energy consumption of an EV, optimizing the routing and the charging strategy using machine learning techniques, graph theory and mathematical optimization. Han et al. have used energy consumption prediction, powertrain modeling and MILP to develop an integrated EV charging and routing system [1].

B. PROBLEM STATEMENT

EV drivers might find it difficult to judge whether the energy stored in the battery is enough to carry out a journey they have planned since energy use depends on the vehicle, the battery, the distance to be travelled, the speed, the road, the weather and the driving conditions. Neither the distance nor the travel time will show what amount of energy will be needed or what the battery level will be when they arrive. When making a decision to charge, due account must be taken of the location of the station, the accessibility of the route, the remaining battery energy, the amount of charging, the charging time and the cost. The system should therefore combine route generation, energy prediction, SOC estimation, identification of the charging station and charging optimisation into a single workflow.

C. OBJECTIVES

- Develop a Chennai-focused EV trip-planning application using real geographical locations and a road-network graph.
- Predict EV trip energy consumption using XGBoost with vehicle, battery, route, speed and environmental features.
- Estimate driving range and arrival battery SOC.
- Generate primary and alternative routes using graph-based processing and Dijkstra's algorithm.
- Identify charging stations when the predicted trip cannot maintain the minimum 25% SOC reserve.
- Optimize charging requirements using a PuLP-based mixed-integer linear programming model.

D. CONTRIBUTIONS OF THIS WORK

The designed system brings together location processing, route creation, energy prediction using XGBoost, SOC and range estimation, route-aware charging station filtering, and MILP charging using PuLP into one Chennai-oriented web application.

E. SCOPE OF THE PROJECT

System design for electric vehicle trips is based on the area of Chennai District. System users enter their starting point, destination, and EV type along with the battery state of charge. System performs routing, estimates energy consumption during the trip with XGBoost algorithm, computes the SOC and range of the vehicle upon reaching the destination, and checks for a 25% minimum SOC buffer. In case it is impossible to keep SOC buffer, system finds available charging stations and charges them optimally via PuLP/MILP.

II. RELATED WORK

Trip planning for EVs that have been proposed by existing researchers includes the areas of energy prediction, route planning, charging station recommendation and charging optimization using Machine Learning, Graph Algorithms and Mathematical Optimization methods. In [1], Han et al. included the concepts of real-time energy prediction, EV powertrain and MILP in their EV charging and routing. In [2], Al-Zuhairi et al. have included the concepts of EV energy-consumption prediction and charging station recommendation using federated learning. In [3], Hussain et al. assessed machine learning and ensemble approaches for energy prediction of EVs and in [4], Huang et al. included the concept of sequence-aware energy prediction using accessible pre-trip data. All of the above-mentioned researchers have focused on certain or all the aspects of trip planning for EVs as can be seen in Table I. The proposed approach incorporates route selection, XGBoost energy prediction, SOC and range calculation, route-based filtering of charging stations and MILP based on PuLP for charging optimization in a web application for Chennai.

TABLE I. COMPARATIVE ANALYSIS OF RELATED EV ENERGY AND CHARGING SYSTEMS

Study	Energy Prediction	Routing / Charging	Primary Focus
Han et al. [1]	LSTM + EVPM	Routing + MILP	Energy uncertainty, routing and charging
Al-Zuhairi et al. [2]	Federated ML	Routing + station recommendation	Energy prediction and station recommendation
Hussain et al. [3]	ML / ensemble	Limited	Energy-consumption prediction
Huang et al. [4]	Sequence-aware DL	Pre-trip data	Accessible pre-trip prediction
Boubaker et al. [5]	Load forecasting	Dijkstra + station allocation	Charging-station allocation

RESEARCH GAP

The existing solutions concentrate more on energy prediction, charging stations suggestions, or route planning and optimization. The present research combines all these aspects into one trip planning process along with location processing, route selection, energy prediction, and charging optimization for Chennai.

III. PROPOSED SYSTEM AND METHODOLOGY

A. SYSTEM ARCHITECTURE

The proposed framework combines location calculation, route generation, EV energy forecasting, SOC and range estimation, charging station selection, and optimal charging operations. The web interface is built based on React technology for frontend, FastAPI and Python for backend calculation, and MongoDB for database management. Origin and destination chosen by the user are then used to develop primary and alternative routes using Dijkstra's algorithm. The selected route is provided to XGBoost for energy forecasting and SOC and range estimation. In case SOC reserve of 25% cannot be guaranteed, reachable charging stations are determined and processed through MILP optimization using PuLP.

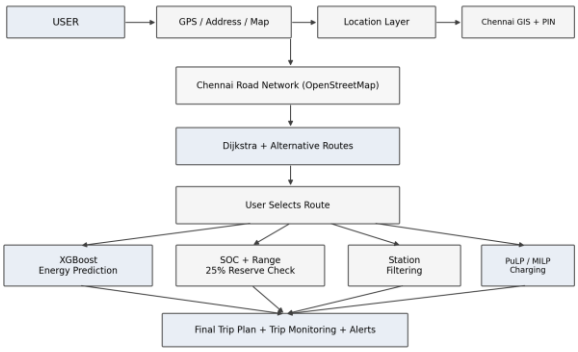


Fig.1. Overall architecture of the proposed AI-assisted EV range prediction and smart charging optimization system.

B. GPS AND ADDRESS RESOLUTION

The system derives the source and destination coordinates through web browser location services, addresses or landmarks searches, or the map itself. The chosen coordinates will be filtered via the Chennai location resolution component using GCC GIS and postal boundary information.

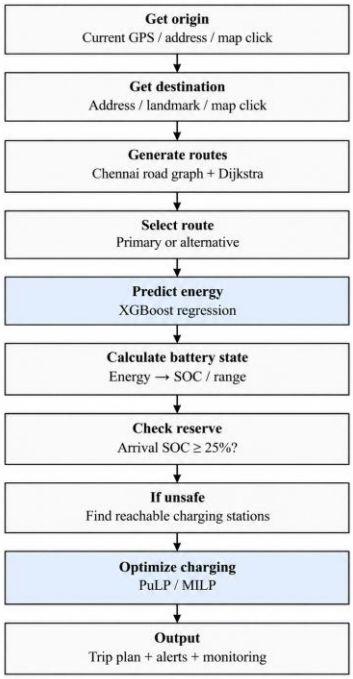


Fig.2. End-to-end workflow of the proposed EV trip-planning system.

C. CHENNAI GIS AND ROAD-NETWORK PROCESSING

The routes that will be created by the system include both the main path and some other possible paths between the specified origin and destination. The algorithm to calculate the main path uses Dijkstra's shortest path, while the other

possible routes are calculated using a k-shortest path algorithm. Comparison of routes will be based on distance and predicted travel time.

D. ROUTE GENERATION AND ALTERNATIVE SELECTION

The primary route is generated using Dijkstra's shortest-path algorithm. The road network is represented as:

$$G = (V, E)$$

where V is the set of nodes and E is the set of road edges.
Let s represent the origin node and d represent the destination node. The primary route is obtained as:

$$R_{primary} = Dijkstra(G, s, d)$$

where the edge weight represents travel time.

$$d(v) = \min_{u \in N(v)} \{d(u) + w(u, v)\}$$

where $d(v)$ is the shortest known path cost to node v, $N(v)$ represents the nodes connected to v ,and $w(u, v)$ represents the travel-time weight of the road segment between nodes u and v.
The alternative paths are obtained through the k-shortest path algorithm. The paths are then compared based on distance and estimated travel time so that the user can choose his or her preferred path for further analysis.

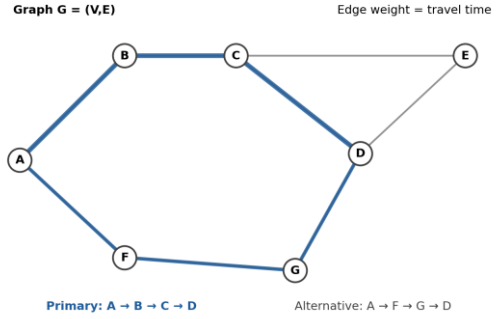


Fig.3 Graph-based route generation using Dijkstra's shortest-path algorithm and alternative-path processing.

E. EV INPUT PARAMETERS

The input parameters include EV make and model, battery capacity and type, motor capacity, mass of the vehicle, initial state of charge, average speed, state of health, ambient temperature, weather condition, road condition, and route distance.

F. XGBOOST ENERGY PREDICTION

The problem of energy consumption is regarded as a supervised regression task. The input variables contain information on the vehicle, the trip itself, as well as environmental and driving factors. While categorical input variables are one-hot encoded, numeric input variables are used as they are. A vehicle-aware splitting of the dataset into training and test sets is carried out, which ensures that no trip belonging to the same vehicle is present in both datasets. Eighty percent of the data is used for training, and twenty percent for testing, yielding 16,050 training samples and 3,950 test samples.
XGBoost regression is represented as

$$E_{pred} = f_{XGBoost}(X)$$

where X represents the processed feature vector and E_{pred} represents the predicted trip energy consumption in kWh.

This model contains 300 estimators with tree depth 6, learning rate 0.05, subsampling 0.9, and feature subsampling 0.9. Five regression models including Random Forest, Extra Trees, Gradient Boosting, XGBoost, and CatBoost have been considered.

TABLE III-COMPARISON OF MACHINE LEARNING MODELS FOR ENERGY PREDICTION

Model	Test R^2	Test MAE	Test RMSE	CV R^2 Mean	CV R^2 Std
Extra Trees	0.9975	0.1930	0.3838	0.9969	0.0009
XGBoost	**0.9978**	**0.1481**	**0.3613**	0.9969	0.0009
CatBoost	0.9972	0.1744	0.4044	0.9969	0.0012
Gradient Boosting	0.9969	0.2694	0.4243	0.9967	0.0002
Random Forest	0.9947	0.3019	0.5605	0.9940	0.0010

XGBoost obtained a test R^2 of 0.9978, MAE of 0.1481 and RMSE of 0.3613 and was therefore selected for energy prediction.

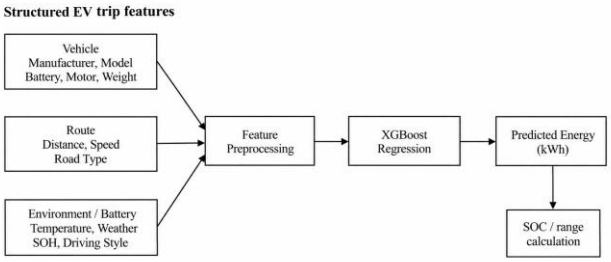


Fig.4. XGBoost-based EV energy-consumption prediction pipeline.

G. SOC AND RANGE ESTIMATION

The available battery energy at departure is calculated as:

$$E_{available} = C_{battery} \times \frac{SOC_{initial}}{100}$$

where $C_{battery}$ is battery capacity in kWh.

The remaining battery energy after the predicted journey is:

$$E_{remaining} = \max(0, E_{available} - E_{pred})$$

The expected arrival SOC is calculated as:

$$SOC_{arrival} = \frac{E_{remaining}}{C_{battery}} \times 100$$

A minimum reserve condition is imposed:

$$SOC_{arrival} \geq SOC_{reserve}$$

where:

$$SOC_{reserve} = 25\%$$

Predicted energy is also translated into predicted range based on the usable energy and consumption along the route that was predicted. The 25% reserve is only a planning restriction and should not be thought to be an actual BMS reading

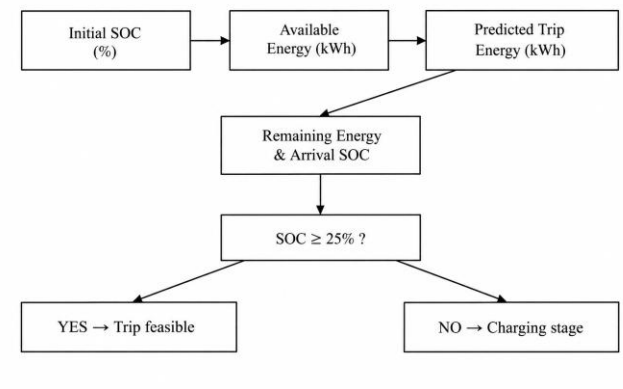


Fig. 5. Battery-energy, SOC, and reserve-feasibility estimation process.

H. LONG-DISTANCE PREDICTION HANDLING

The data set is restricted to a certain range of distance of trips. In case of trips which are beyond the range of observations, the model makes use of controlled energy scaling using a benchmarking prediction and guardrail on efficiency.

I. CHARGING-STATION IDENTIFICATION

Where the estimated journey fails to keep the 25% reserve SOC, charging station processing is started. The stations are selected depending on how close they are to the route, their positioning in relation to the journey, and whether they can be reached while not breaking the 25% reserve.

IV. CHARGING PLANNING AND OPTIMIZATION

A. CHARGING DECISION LOGIC

The charge plan will work in cases where it is not possible to complete the selected trip without violating the minimum SOC requirement. Feasible solutions will be generated through route-aware station processing that will pass on to MILP optimization.

B. OPTIMIZATION VARIABLES AND OBJECTIVE

Let x_i be a binary variable indicating whether charging station i is selected:

$$x_i \in \{0, 1\}$$

Let q_i denote the amount of energy charged at station i.

The optimization minimizes charging cost and charging time:

$$\min Z = w_c C_{charge} + w_t T_{charge}$$

where w_c and w_t are the charging-cost and charging-time weights. The current implementation uses a cost weight of 0.6 and a time weight of 0.4

The reserve constraint is represented as:

$$SOC_{arrival} \geq 25\%$$

Additional constraints ensure that charging occurs only at selected candidate stations and that charging quantities remain within battery capacity and route-energy requirements. When verified charging-power or cost data are unavailable, prototype uses assumed values for these parameters.

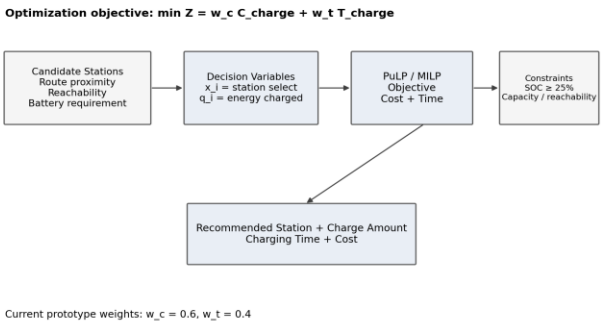


Fig. 6. PuLP-based mixed-integer linear programming framework for charging decision optimization.

V. SYSTEM IMPLEMENTATION

A. SOFTWARE ARCHITECTURE

The software application utilizes a client-server approach. The React and TypeScript frameworks deliver the user interface, while MapLibre GL JS provides the map visualizations. The back-end services use the FastAPI and Python frameworks. OSMnx and NetworkX provide road network analysis and Dijkstra routing. The MongoDB database manages the project geospatial data. The XGBoost

framework delivers energy predictions, while the PuLP/MILP framework manages charging optimization.

TABLE II. SOFTWARE AND SYSTEM COMPONENTS

Component	Technology	Purpose
Frontend	React + TypeScript	Interactive trip-planning interface
Mapping	MapLibre GL JS	Map display and route visualization
Backend	FastAPI + Python	API and application services
Routing	OSMnx + NetworkX	Chennai road graph and Dijkstra routing
Database	MongoDB	Document storage and geospatial indexing
Prediction	XGBoost	EV energy-consumption prediction
Optimization	PuLP / MILP	Charging decision optimization

B. WEB APPLICATION IMPLEMENTATION

The web application allows users to select the origin and destination, choose an EV model and provide initial SOC and operating conditions. It displays route options, predicted energy, estimated range, arrival SOC and charging recommendations.

C. GEOSPATIAL DATA LAYER

OpenStreetMap data is used to construct the Chennai road network, while GCC GIS records and postal PIN polygons support geographic location resolution.

D. DATASET PREPARATION

The dataset contains 20,000 structured EV trip records with vehicle, battery, route, speed, environmental and driving-condition features. After data cleaning, a vehicle-aware 80/20 split produced 16,050 training and 3,950 testing records. Five-fold GroupKFold validation was used for model comparison.

VI. EXPERIMENTAL SETUP

A. MACHINE-LEARNING EVALUATION

The models are evaluated using MAE, RMSE and R^2 where lower MAE and RMSE and higher R^2 indicate better prediction performance.

B. ROUTE EVALUATION

TABLE IV. ROUTE ALTERNATIVE EVALUATION

Trip	Route	Distance (km)	Duration (min)
Sachidanandam Lane, Kosapet, Chennai, Tamil Nadu, 600012 **To** **G.S.T. ROAD, MEENAMPAKKAM, Chennai, Tamil Nadu, 600027**	Primary	20.05	25
	Alternative 1	23.15	34
	Alternative 2	28.18	47

C. TRIP-LEVEL EV EVALUATION

For each selected route, the system records initial SOC, battery capacity, route distance, predicted energy, estimated range, safe range, arrival SOC, charging requirement and selected charging station where applicable.

TABLE V. TRIP-LEVEL ENERGY AND SOC EVALUATION

Initial SOC	Trip Energy (kWh)	Range to 0% (km)	Safe Range (km)	Arrival SOC
50%	3.51	172.6	86.3	44.2%
80%	3.51	310.6	224.4	84.2%

D. CHARGING OPTIMIZATION EVALUATION

TABLE VI. CHARGING OPTIMIZATION RESULTS

Scenario	SOC	Energy	Arrival SOC	Decision
S1	28%	3.51 kWh	22.2%	MILP charging
S2	20%	3.51 kWh	14.2%	No feasible station
S3	20%	3.84 kWh	13.7%	No feasible station

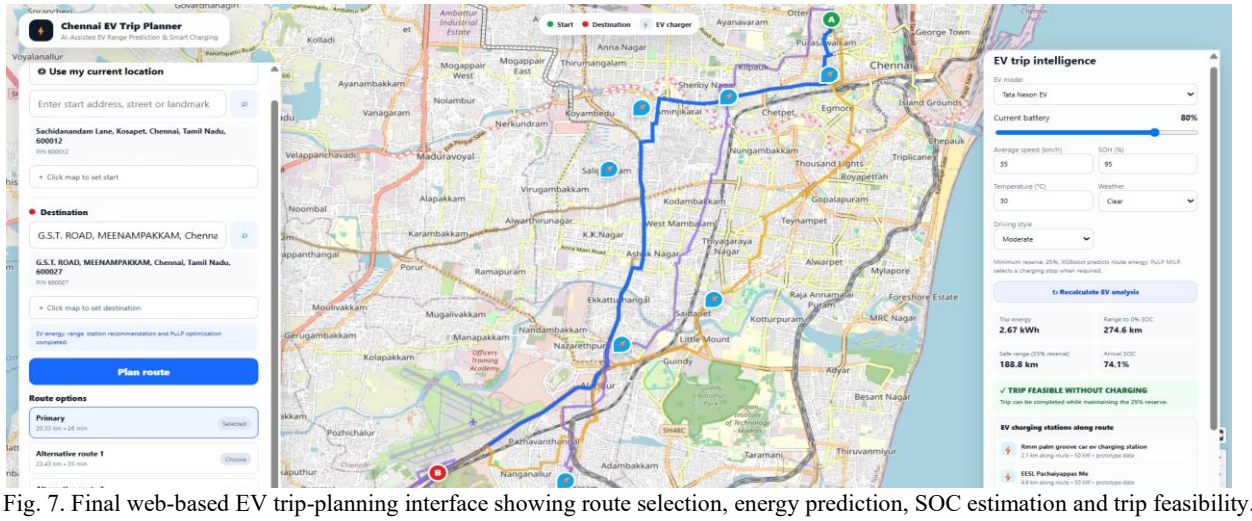


Fig. 7. Final web-based EV trip-planning interface showing route selection, energy prediction, SOC estimation and trip feasibility.

The charging scenarios demonstrate the behaviour of the reserve-constrained planning process. In S1, the initial SOC of 28% results in an arrival SOC of 22.2%, triggering MILP charging because the 25% reserve is not maintained. In S2 and S3, the arrival SOC is 14.2% and 13.7%, respectively, and no feasible station is found under the considered conditions.

E. COMPLETE TRIP EVALUATION

TABLE VII. END-TO-END TRIP EVALUATION

Parameter	Result
EV model	BYD Atto 3
Initial SOC	80%
Primary distance	14.77 km
Duration	22 min
Predicted trip energy	2.55 kWh
Safe range (25% reserve)	192.9 km
Arrival SOC	75.8%
Charging required	No

The end-to-end evaluation shows that the chosen trip is possible without any cost with an estimated final state of charge of 75.8%.

VII. RESULTS AND DISCUSSION

A. ENERGY PREDICTION RESULTS

XGBoost achieved the best test performance with $R^2 = 0.9978$,MAE of 0.1481 and RMSE of 0.3613. The grouped validation results also support its selection for energy prediction. However, these results are based on the project dataset rather than independently collected fleet telemetry.

B. ROUTE PLANNING RESULTS

The router module creates the Dijkstra route along with alternate routes. As presented in Table IV, the system is able to provide alternate routes with varying distance and time values before conducting energy analysis.

C. SOC AND RANGE RESULTS

Energy calculated from prediction is added to the starting SOC to compute energy remaining, arrival SOC, and range. The 25 percent reserve is used to determine if the trip is feasible; trips below this level move on to the charging station evaluation phase.

D. CHARGING OPTIMIZATION RESULTS

However, in case where charging is needed, the available charging stations are then fed into the PuLP optimizer, taking into account the cost, time, and energy constraints for charging. Table VI provides an example of reserve-constrained charging decisions. The charging rates and power levels used in the prototype are planning estimates.

E. INTEGRATED SYSTEM RESULTS

The entire process takes in consideration the route selection, energy calculation, SOC/Range calculation, station selection and charging optimization in one planning process.

VIII. LIMITATIONS

The model is trained on the project's available dataset rather than an independent fleet-telemetry dataset, thus the accuracy of the results presented cannot be guaranteed in the real world. The algorithm relies on the OpenStreetMap and GCC GIS database, but there is no support for real-time traffic data. The charging power and cost will assume prototype values where no verification from the stations' actual data is provided. The 25% of SOC buffer zone is the design constraint and cannot replace the BMS of the vehicle.

IX. CONCLUSION AND FUTURE WORK

This paper presented an AI-assisted EV trip-planning system integrating Chennai road-network routing, XGBoost energy prediction, SOC and range estimation, charging-station filtering and PuLP-based MILP optimization. XGBoost achieved $R^2 = 0.9978$, MAE value of 0.1481, and RMSE value of 0.3613.

In future, attention will be paid to larger real-life EV telematics data, traffic data, validation of charging stations, dynamic charging costs, better battery modeling, and energy-based routing.

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